



Muralidharan D

Mechatronics Engineering Student • UAV & Robotics Enthusiast

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EDUCATION

B.E. Mechatronics Engg.
Kumaraguru College of Technology
2023 – 2027 | CGPA: 7.5

Higher Secondary (12th)
Kongu Kalvi Matric HR Sec School
2022 – 2023 | 86%

SKILLS

UAV & Drones
Assembly • Edge AI • Piloting

Design & Modeling
Fusion 360 • Blender • 3D Print

PCB Design
KiCad • EasyEDA • THT/SMD

Programming
Arduino C • Python

AI / Vision
YOLO Image Processing • RVC

Manufacturing
Welding • Cutting • Fabrication •
Soldering • EV Assembly

Other
VR Dev (Unity XR) • PC Servicing

LANGUAGES

Tamil
Native

English
Professional Proficiency

LEADERSHIP

President — RAC
2025–26

Embedded Lead — RAC
2024–25

Tech Support Exec.
Mechatronics Dept. • 2024–25

Visual Design Lead
Aeromodelling Club • 2024–25

Technical Member
Aeromodelling Club • 2023–24

OBJECTIVE

Motivated Mechatronics Engineering student at KCT with hands-on expertise in UAV systems, embedded electronics, robotics, and VR-based industrial training. Seeking an internship in the UAV / Robotics domain to contribute in 3D modeling, PCB design, AI model development, and hardware prototyping.

EXPERIENCE

Product Developer • PROTOSEM Innovation Engineers 2024 – Present

Product-based innovation program (equivalent to internship) • KCT, Coimbatore

- Developing FORGE — VR-based industrial maintenance training system for Meta Quest 3
- Designed 407+ object hydraulic cylinder workshop in Blender with PBR materials
- Implementing XR interactions in Unity using Meta XR SDK and OpenXR

PROJECTS

FORGE — VR-Based Industrial Training System • 2024 – Present

- Photorealistic hydraulic cylinder servicing workshop in Blender (407+ objects, PBR)
- XR interactions for Meta Quest 3 using Unity XR Toolkit and OpenXR

Role: Hustler — Blender environment, Unity VR interactions & scripting

Disaster Management Drone • 2024 – 2025

- Multi-role UAV with Edge AI target recognition and disaster payload drop capability
- AIR 3 in Phase-1 technical report at Aerothon 2025 among 89 national teams

Role: Embedded systems, AI integration & mechanical design

Delivery Drone • 2024

- End-to-end autonomous delivery UAV — frame design, flight controller tuning, payload mechanism

Role: Team Captain — overseeing complete project lifecycle

Agriculture Drone • 2024

- Precision agriculture UAV with variable-rate spraying and GPS-based field mapping

ACHIEVEMENTS

🏆 **Aerothon 2025 — AIR 3 & Special Recognition** 2025

All-India Rank 3 in Phase-1 technical report among 89 teams • National finals at Christ University, Bangalore • Special recognition for OICC & Disaster Payload Drop

🏆 **Smart India Hackathon (SIH) 2025 — National Finalist** 2025

Top 6 finalist team for Namami Gange Smart River Water Quality UAV Surveillance System • Role: Vice Captain

🏆 **Leadership & Volunteerism Award — KCT RIDE Awards** May 2025

Recognized for outstanding contribution to Robotics & Automation Club (2024–25)

🏆 **1st Place — Ideathon, Aeromodelling Club KCT** December 2024

Best UAV Design award for VTOL Long Endurance concept — design & concept lead

🏆 **Runner-Up — ROBO MANIA (Speed Trial & Tug of War)** February 2025

Secured runner-up in both events at KCT as Team Captain